

Graph-Guided Latent Diffusion for Scalable Multi-Agent Path Finding in Dynamic Environments

Elena Rostova¹[0000-0002-1825-0097], Marcus Vance^{1,2}[0000-0001-2345-6789],
Priya Sharma²[0000-0002-9876-5432], and Arthur
Pendelton³[0000-0003-4567-8901]

¹ Nexa AI Research Institute, San Francisco, CA 94105, USA
e.rostova@nexa-research.org, m.vance@nexa-research.org

² Synapse Center for Autonomous Systems, Cambridge CB2 1PZ, UK
p.sharma@synapse-institute.org

³ Meridian Quantum Systems, 8001 Zurich, Switzerland
a.pendelton@meridian-qs.ch

Abstract. Multi-Agent Path Finding (MAPF) in large-scale dynamic environments presents severe computational challenges, particularly when balancing non-collision guarantees with trajectory optimality under strict real-time deadlines. Traditional search-based algorithms struggle to scale when hundreds of autonomous agents navigate densely packed domains, whereas pure reinforcement learning models frequently suffer from unstable convergence and horizon limits. In this paper, we introduce *Graph-Guided Latent Diffusion* (GGLD), a framework that formulates multi-agent trajectory synthesis as a conditioned continuous-space diffusion process guided by dynamic spatial-temporal graph representations. GGLD encodes spatial topologies and dynamic agent inter-dependencies into a low-dimensional latent space using spatial-temporal graph attention networks. A probabilistic diffusion model then refines candidate trajectory manifolds while enforcing hard non-collision boundaries via energy-guided Langevin dynamics. Extensive empirical evaluations across benchmark warehouse topologies demonstrate that GGLD achieves up to a 38.4% reduction in total travel time and improves path completion rates by 24.2% over state-of-the-art baselines, while maintaining real-time decision inference under 15 ms per timestep.

Keywords: Multi-Agent Path Finding · Latent Diffusion Models · Graph Neural Networks · Dynamic Trajectory Optimization · Autonomous Robotics.

1 Introduction

Multi-Agent Path Finding (MAPF) is a core algorithmic problem in modern automated logistics, warehouse robotics, and autonomous transportation systems.